

The balance reading is given by $17g > 17m$ i.e. $> 50\text{ g}$

9 A At the pole, $g = \frac{GM}{R^2}$ -----(1)

At height h , $0.99g = \frac{GM}{(R+h)^2}$ -----(2)

$$\frac{(2)}{(1)} \quad 0.99 = \frac{R^2}{(R+h)^2}$$

$$R+h = \frac{R}{\sqrt{0.99}}$$

$$h = \left(\frac{1}{\sqrt{0.99}} - 1 \right) (6400) = 32 \text{ km}$$

10 B For a satellite of mass m in orbit at distance r from Earth of mass M